Machine Learning Methods for Identifying Lung Cancer

Temirlan Nauatbek[2301], Ulan Kalken[2303], Vlad Obedkov[2303]

Big Data Analysis, Astana IT University, Astana, Kazakhstan.

Abstract

The worldwide death toll from lung cancer continues to rise because this disease accounts for 20 percent of all cancer-related fatalities. The survival rates of patients improve when doctors detect cancer at an early stage but current screening methods including LDCT and chest radiographs produce variable results and generate numerous false alarms and face challenges with large-scale implementation. The development of machine learning (ML) and deep learning (DL) technology has led to the creation of automated systems which detect pulmonary nodules at levels comparable to those of trained radiologists.

The research implements ML and DL methods for lung cancer detection through analysis of three public datasets including LIDC-IDRI and LUNA16 and Kaggle clinical data to achieve reproducibility. The evaluation of ResNet50 convolutional neural network against classical algorithms (SVM, Random Forest, KNN) assesses their performance through accuracy and sensitivity and specificity and F1-score and AUC-ROC metrics. The research shows that deep learning models achieve superior results than traditional methods in both data generalization and feature detection capabilities. The study demonstrates how clinical practice benefits from implementing CAD tools that use data-driven methods and maintain patient privacy to decrease medical workloads and enhance early cancer detection results. The research demonstrates that deep learning systems have the ability to improve precision oncology while decreasing worldwide lung cancer deaths.

1 Introduction

1.1 Background and Context

Lung cancer remains one of the most dangerous forms of cancer, consistently ranking among the top causes of cancer-related deaths. According to global statistics, it

accounts for nearly 18 percents of all cancer mortality every year[1]. Despite noticeable progress in diagnostic imaging and treatment strategies, a large portion of cases is still identified at later stages, when therapy is more complex and outcomes are significantly worse. Early detection remains one of the strongest predictors of survival, yet achieving it reliably on a population scale is still challenging.

Low-dose computed tomography (LDCT) has become a central tool for screening high-risk individuals. The National Lung Screening Trial (NLST) demonstrated that LDCT screening reduces lung cancer mortality compared to chest X-rays[2]. However, LDCT brings its own limitations: radiologists must review a large number of slices per patient, which increases workload; many findings are ambiguous; and false positives are common, leading to unnecessary follow-up procedures. Inter-reader variability also remains a problem—radiologists may interpret the same CT scan differently, especially when nodules are small or atypical.

Against this backdrop, artificial intelligence has emerged as an important support tool. Machine learning (ML) and especially deep learning (DL) have shown the capacity to analyze CT scans efficiently and consistently. With the increasing availability of annotated datasets, such as LIDC-IDRI and curated challenges like LUNA16, researchers have been able to experiment with a range of algorithms. Convolutional neural networks (CNNs) are particularly well suited for this kind of visual pattern recognition. They can identify subtle features—variations in texture, edges, shapes, and spatial contexts—that are often difficult to capture using traditional hand-designed descriptors[3, 4].

Some landmark works have demonstrated what DL models are capable of. For instance, the 3D CNN created by Ardila et al. achieved radiologist-level AUC on the NLST dataset, suggesting that automated systems may realistically support large-scale screening[3]. Benchmarks like the LUNA16 challenge helped unify evaluation practices by offering standardized datasets and clear performance metrics[5]. Reviews and meta-analyses further strengthen the case for DL approaches: studies consistently show that CNNs tend to outperform older ML models, particularly in tasks involving nodule detection, malignancy estimation, and triaging of suspicious lesions[6, 7].

In short, AI has increasingly become a core component of modern lung cancer imaging research. Yet, while progress has been significant, there are still practical obstacles and open questions that motivate further study.

1.2 Problem Statement and Research Gap

Although deep learning has shown impressive performance in controlled experiments, several challenges limit the reliability and routine use of such models in clinical environments.

One major issue is dataset heterogeneity. CT scans vary widely across medical centers due to differences in hardware, scanning protocols, reconstruction algorithms, and patient populations. These variations create domain shifts that can reduce the performance of models when applied outside the environment they were trained in[8]. A network that performs well on one dataset may struggle with images that have different texture distributions or noise patterns.

Another challenge is the scarcity of high-quality labeled data. Annotating lung nodules in three-dimensional CT volumes is time-consuming and requires skilled radiologists. Even large datasets like LIDC-IDRI[9], although widely used, include varying interpretations: nodules were annotated by multiple radiologists independently, leading to differences in boundaries and malignancy scores. As a result, researchers often face inconsistent labels, which complicates training and model evaluation.

A third concern relates to interpretability. CNNs often produce accurate results without offering much explanation. For real-world medical use, clinicians must be able to justify decisions, particularly for high-stakes diagnoses like cancer. Without transparency, even high-performing models may be treated with caution.

Privacy and ethical constraints also shape how medical imaging datasets can be collected and shared. Regulations like GDPR and HIPAA limit centralization of patient data, making it harder to assemble diverse datasets. These constraints have pushed research toward alternative approaches such as federated and semi-supervised learning, where data do not leave the source institution but still contribute to a shared model[10].

While many studies address individual aspects—generalization, transfer learning, radiomics vs. DL, preprocessing, interpretability—comparisons between classical ML and DL under consistent conditions remain limited. Similarly, the specific role of preprocessing (e.g., resampling, normalization, segmentation) and augmentation (e.g., flips, rotations, intensity adjustments) is not fully understood, even though these steps can greatly influence model performance and stability[4].

Thus, the motivation for this study is to examine classical ML and DL methods side by side, using the same datasets, comparable evaluation metrics, and consistent preprocessing settings.

1.3 Research Objectives and Questions

This research aims to systematically study how classical ML techniques and deep learning models perform in detecting lung cancer when applied to CT-based datasets and structured clinical data. The goal is not simply to show that deep learning performs better—which is widely known—but to understand how these methods compare under shared experimental conditions and what factors influence their performance.

The study investigates four main questions:

1. How do classical ML algorithms compare with a modern CNN-based architecture? We evaluate SVM, Random Forest, and KNN against ResNet50 to understand the advantage of deep learned features over handcrafted radiomic ones.

2. What impact do preprocessing and data augmentation steps have on the final performance? Since CT data can vary substantially in resolution, noise, and nodule characteristics, these steps may play a crucial role in ensuring stable, generalizable results. This also provides insight into how sensitive different algorithms are to data quality.

3. How effective is transfer learning for this task? Medical datasets are smaller than typical computer vision datasets, making transfer learning especially valuable. The study evaluates whether ImageNet pretrained weights improve performance and training efficiency.

4. What practical issues must be considered when integrating ML into real clinical workflows? This includes interpretability, domain adaptation, privacy constraints, and clinician trust. Addressing these considerations is essential for the realistic deployment of AI tools.

To answer these questions, the study uses three open datasets: LIDC-IDRI[11], LUNA16[12], and a Kaggle lung cancer prediction dataset[13]. Performance is measured using Accuracy, Sensitivity, Specificity, F1-score, and AUC-ROC—metrics that are commonly reported in medical AI research.

1.4 Significance and Contributions

This study contributes to the growing literature on AI-driven diagnostic imaging in several ways.

First, it provides a direct comparison between classical machine learning and deep learning using consistent experimental conditions. By training all models on the same datasets and evaluating them with identical metrics, the study offers a fair assessment of where each approach excels and where it falls short.

Second, the work highlights how preprocessing and augmentation influence results, something that is often overlooked. Understanding these factors is crucial for building reliable diagnostic systems, especially when datasets originate from diverse sources.

Third, by using publicly available datasets, the study promotes reproducibility—a key requirement for modern research[9, 14]. Anyone can replicate the pipeline and verify the findings.

Finally, the research points toward practical considerations for real-world adoption. The discussion addresses interpretability, dataset bias, privacy concerns, and the potential role of federated and semi-supervised learning, which are likely to shape the next generation of clinical AI systems.

1.5 Structure of the Paper

This work is structured as follows. Section 2 reviews previous work on lung cancer detection, including deep learning architectures, radiomics approaches, and recent developments in transfer and federated learning. Section 3 describes the methodology used in this study, covering the datasets, preprocessing pipeline, feature extraction, model training procedures, and evaluation metrics. Section 4 presents the experimental results for both classical machine learning models and the ResNet50 deep learning model. Section 5 provides a discussion of these findings, highlighting their implications, limitations, and connections to existing research. Finally, Section 6 concludes the work and outlines potential directions for future studies.

2 Literature Review

2.1 Deep Learning Architectures for Nodule Detection

Deep learning has quickly become central to lung nodule detection research. One of the first well-known works used a 3D convolutional network to predict cancer risk from low-dose CT scans. On the NLST dataset it reached an AUC of 0.944, which is about

the same as experienced radiologists [3]. This showed that an automated system could realistically support population-level screening. In parallel, the LUNA16 challenge provided a shared benchmark for testing algorithms. Results from that project confirmed that ensembles of detectors combined with CNN-based false-positive reduction usually perform better than single models [5].

Later studies focused on improving network design. For example, Zheng et al. built a multi-view CNN using maximum intensity projections (MIP). This approach followed a similar reading pattern to radiologists and proved more sensitive for very small nodules [15]. Zhang et al. explored multi-scale attention, which helped the model detect nodules across different size ranges [16]. Liu et al. introduced the 3DFPN-HS^2 model, which made use of cross-slice information to keep sensitivity high while still holding down false positives [17].

Another important step has been testing these systems in real hospitals. A multi-center study showed that CNN-based CAD tools could work on scans collected under different conditions and even improve radiologist performance when used side by side [18]. Similar work on chest radiographs found that deep learning algorithms sometimes matched or slightly surpassed human accuracy, pointing to CAD as a useful tool beyond CT [19].

Overall, CAD development for lung cancer has moved from early proof-of-concept 3D models, through public challenges like LUNA16, toward more advanced attention and multi-scale methods, and finally into clinical validation. Still, issues such as biased datasets, retrospective evaluations, and heavy computational demands remain, and these continue to limit how widely the systems can be adopted in practice.

2.2 Federated, Transfer, and Semi-Supervised Learning Approaches

Standard supervised deep learning usually needs large labeled datasets, which are often hard to build and even harder to share in healthcare. Because of this, many recent works explore data-efficient and privacy-friendly approaches. One example is self-supervised pretraining: models first learn general features from unlabeled scans and then get fine-tuned on a smaller labeled set. This setup improved both accuracy and robustness for nodule detection and also cut down false positives [20]. A related idea, the 3DFPN-HS^2 framework, showed that adding cross-slice context makes detection more stable and helps balance sensitivity and specificity [17].

Federated learning (FL) is another direction. In FL, hospitals train models together but keep their data locally. The FedDUS framework tested this idea with a semi-supervised setup and reported better generalization across centers than either local-only or fully centralized training [10]. Reviews of FL and transfer learning in oncology also point out that these methods protect privacy and speed up convergence when datasets are limited [21]. Some authors expect that mixed FL and TL schemes will become more common in the near future.

A commentary in *Nature Digital Medicine* took a broader look across breast, lung, and prostate cancer. It noted that FL is not only about privacy but also about enabling collaboration between hospitals, which is critical for scaling AI to real clinical use [22].

Overall, these works show that progress in lung cancer detection is not just about higher accuracy. It also depends on how data are shared and used. Semi-supervised training, federated setups, and transfer learning offer ways to move from isolated research models to systems that can actually work in practice.

2.3 Radiomics and Clinical Applications

Besides detecting nodules, ML methods are used in radiomics to extract imaging biomarkers. Several studies showed that CT or PET/CT radiomics can predict EGFR mutation status in NSCLC patients using ML classifiers. These approaches offer a non-invasive option compared to biopsy and nomograms that combine radiomic and clinical features can further improve prediction [23].

Systematic reviews also describe the growing role of radiomics in prognosis and therapy. A PRISMA-based review of more than 40 studies reported that radiomic signatures can predict survival, treatment response, and even radiation toxicity. Models that mix imaging, clinical, and dose features usually perform best [24]. An educational review walked through the whole pipeline, from image collection and segmentation to modeling, and underlined the need for reproducibility and standards [14].

More recent reviews focus on deep learning–based radiomics, where CNNs, autoencoders, and transformers take the place of handcrafted features. These can capture richer patterns but also raise questions of interpretability and cross-center consistency. A 2025 paper stressed the importance of multimodal fusion (CT, MRI, PET, genomics) and explainable AI, along with quality control, to make radiomics useful in practice [25].

In sum, radiomics expands ML from simple detection to prognosis and treatment planning. The main obstacles are still the lack of standardization, different imaging protocols, and limited prospective validation. Large multicenter studies and harmonized workflows will be needed before radiomics can become routine.

2.4 Review and Meta-Analyses

Besides individual studies, there are also many reviews and meta-analyses that look at the bigger picture. A systematic review from 2023 covered deep learning methods for CT screening. It described the move away from handcrafted features toward CNN and U-Net pipelines, often combined with clinical data. The same paper also stressed that stronger external validation is still needed [4].

A separate meta-analysis showed that AI systems usually increase sensitivity and specificity compared to traditional screening. At the same time, it found that dataset quality and diversity play a major role in outcomes, which helps explain why many models still struggle to generalize [8].

Some broader surveys give an even wider perspective. One review listed 3D CNNs, radiomics + DL hybrids, and transfer learning as leading strategies. However, it also pointed out that most studies lack prospective trials and standardized reporting [26]. A high-level review in *Nature* discussed emerging themes such as multimodal fusion, transformers, and explainable AI. It also raised ethical concerns, including bias and the absence of reimbursement structures [27].

In addition, a Springer review from 2024 presented a taxonomy of methods. It outlined common preprocessing steps, segmentation approaches, feature extraction techniques, and classification models, and also summarized which metrics are most often reported. This kind of structured overview is useful for new researchers entering the field [28].

3 Methods and Materials

3.1 Research Design

This study follows a quantitative experimental research design aimed at developing and evaluating machine learning models for the detection of lung cancer based on computed tomography (CT) images. A quantitative approach is appropriate because the research problem involves measurable imaging data, numerical feature representations, and objective model performance assessment through statistical metrics. The experimental nature of the study ensures that models are tested under controlled conditions, allowing reproducibility and comparison across different machine learning techniques [3, 5]. Supervised learning is employed, where CT scans labeled as benign or malignant are used to train classification algorithms.

The experiment compares classical machine learning algorithms with a deep learning–based approach to assess the relative strengths and weaknesses of each category. Three baseline traditional models are utilized: Support Vector Machine (SVM), Random Forest (RF), and k-Nearest Neighbors (KNN). These algorithms are widely used in medical imaging research due to their robustness and well-established theoretical foundations [21]. SVM is particularly suitable for binary classification tasks with clear decision boundaries, RF handles heterogeneous features and reduces overfitting through ensemble learning, while KNN serves as a distance-based classifier that provides an interpretable baseline.

In addition to classical models, a deep learning model based on the ResNet50 convolutional neural network architecture is included to investigate automatic feature extraction capabilities. Unlike traditional algorithms, which depend on handcrafted features, ResNet50 learns high-level image representations directly from pixel data through hierarchical convolutional operations [19]. Transfer learning is utilized by initializing the network with ImageNet pretrained weights, which accelerates convergence and improves generalization when working with limited medical datasets, as demonstrated in prior research [16].

The study follows a structured pipeline consisting of four phases: dataset acquisition, preprocessing, model training, and evaluation. Preprocessing includes image normalization, resizing, and augmentation, which improve data consistency and reduce overfitting, consistent with best practices in lung nodule detection research [15]. Classical methods rely on radiomics-based feature extraction, while ResNet50 processes raw image inputs. The dataset is split into training and testing subsets using a stratified 80/20 strategy to preserve class balance. Hyperparameter tuning is performed systematically through grid search for classical models and fine-tuned optimization strategies for deep learning.

This experimental design provides a systematic framework to test the hypothesis that machine learning models can enhance lung cancer detection performance compared to conventional methods and supports a fair comparative analysis between traditional and deep learning approaches.

3.2 Data Sources

This work use publicly accessible medical databases to guarantee reproducibility, transparency, and adherence to ethical research norms. Three datasets were chosen due to their pertinence to lung cancer analysis, the presence of expert annotations, and their extensive utilisation in previous machine learning studies.

The primary dataset used is the Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI), which contains 1,018 thoracic CT scans with annotated pulmonary nodules [11]. There are four senior radiologists who individually examined every scan, providing nodule boundary annotations and malignancy likelihood scores. The dataset is heavily utilized as a benchmark for computer-aided detection (CAD) studies for lung cancer due to the fact that it contains comprehensive annotations and the sample size is large. These characteristics render it apt for training supervised classification and detection models.

To simplify preprocessing and ensure consistency with previous work, the LUNA16 dataset is used as a curated subset of LIDC-IDRI [12]. LUNA16 removes ambiguous cases and includes only nodules with a diameter of at least 3 mm. This dataset became widely known through the LUNA16 challenge, which established a standardized evaluation protocol for nodule detection. It also provides predefined train-test splits, which improve experimental comparability across studies [5, 15]. For this study, the official fold-0 split is used to maintain consistency and avoid sampling bias.

In addition to CT-based datasets, dataset from Kaggle titled *Lung Cancer Prediction* is used [13]. Unlike LIDC-IDRI and LUNA16, this dataset consists of structured clinical attributes such as age, smoking history, alcohol consumption, and cancer diagnosis labels. It is included in this study to enable a comparative experiment in which classical machine learning models (SVM, Random Forest, and KNN) are evaluated on non-imaging clinical data. This helps illustrate the practical differences between feature-based and deep learning-based diagnostic approaches and provides additional validation for model robustness across heterogeneous data types.

GENDER	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL CONSUMING	COUGHING	SHORTNESS OF BREATH	SWALLOWING DIFFICULTY	CHEST PAIN	LUNG_CANCER
M	69	1	2	2	1	1	2	1	2	2	2	2	2	2	YES
M	74	2	1	1	1	2	2	2	1	1	1	2	2	2	YES
F	59	1	1	1	2	1	2	1	2	1	2	2	1	2	NO
M	63	2	2	2	1	1	1	1	1	2	1	1	2	2	NO
F	63	1	2	1	1	1	1	1	2	1	2	2	1	1	NO
F	75	1	2	1	1	2	2	2	2	1	2	2	1	1	YES
M	52	2	1	1	1	1	2	1	2	2	2	2	1	2	YES
F	51	2	2	2	2	1	2	2	1	1	1	2	2	1	YES
F	68	2	1	2	1	1	2	1	1	1	1	1	1	1	NO
M	53	2	2	2	2	2	1	2	1	2	1	1	2	2	YES
F	61	2	2	2	2	2	2	1	2	1	2	2	2	1	YES
M	72	1	1	1	1	2	2	2	2	2	2	2	1	2	YES
F	60	2	1	1	1	1	2	1	1	1	1	2	1	1	NO
M	58	2	1	1	1	1	2	2	2	2	2	2	1	2	YES
M	69	2	1	1	1	1	1	2	2	2	2	1	1	2	NO
F	48	1	2	2	2	2	2	2	2	1	2	2	2	1	YES

Fig. 1 Header structure of the Kaggle Lung Cancer Prediction dataset.

Before model training, all data undergo standardized preprocessing. CT images from LIDC-IDRI and LUNA16 are converted from raw scan intensities to Hounsfield Units and resampled to a uniform voxel spacing to reduce variation caused by differences in CT scanners [16]. Lung segmentation is performed to isolate the regions of interest by applying thresholding and morphological operations, following similar methodologies used in previous studies [15]. For convolutional neural network (CNN) training, three-dimensional patches centered on nodule coordinates are extracted and resized to 224×224 pixels. Data augmentation, including rotations, flips, intensity scaling, and Gaussian noise, is applied to reduce overfitting and improve generalization [17].

For classical machine learning models, radiomic features are extracted from segmented nodules using gray-level co-occurrence matrices, texture descriptors, histogram-based statistics, and shape features, consistent with established radiomics practices [23, 24]. To make sure that all numerical attributes contribute equally during model training, feature normalisation using z-score scaling is used. All datasets utilised in this study are anonymised and disseminated publically under open-access licenses, thus obviating the necessity for institutional review board (IRB) permission for their utilisation.

3.3 Evaluation Metrics

Evaluating the performance of machine learning models in medical diagnosis requires reliable and clinically meaningful metrics. Since the task of lung cancer detection is a binary classification problem involving malignant and benign cases, model performance is assessed using metrics derived from the confusion matrix. These metrics are widely used in prior studies on lung cancer detection and provide a foundation for comparing different machine learning approaches [4, 8].

The confusion matrix consists of four outcomes: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Based on these values, Accuracy is calculated as the ratio of correctly classified samples to the total number of samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Although accuracy is a general indicator of correctness, it can be misleading in medical applications where datasets are often imbalanced, meaning malignant cases are significantly fewer than benign ones. In such cases, a model may achieve high accuracy by predicting the majority class only. Therefore, additional metrics are required to ensure that the model performs well on the minority class, which is clinically more important [26].

Sensitivity (also called Recall or True Positive Rate) measures the ability of the model to correctly identify malignant cases, which is crucial in cancer screening where missing a positive case can have serious consequences:

$$Sensitivity = \frac{TP}{TP + FN}$$

Specificity (True Negative Rate) measures how well the model identifies benign cases and helps control false alarms that may lead to unnecessary follow-up examinations:

$$Specificity = \frac{TN}{TN + FP}$$

Precision (Positive Predictive Value) evaluates the proportion of correctly predicted malignant cases among all predicted malignant cases:

$$Precision = \frac{TP}{TP + FP}$$

Since Precision and Recall measure different aspects of performance, the F1-score is used as a balanced metric that combines both:

$$F1\text{-}score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is also reported, as it reflects the model's discriminative ability across different decision thresholds [8]. A higher AUC value indicates better separation between malignant and benign classes, which is essential for early diagnosis. In addition, Precision–Recall AUC (PR-AUC) is considered for imbalanced data settings, as it focuses on the model's performance on the positive class, which is more relevant in cancer detection.

To address class imbalance, Balanced Accuracy is calculated as the average of Sensitivity and Specificity:

$$Balanced\,Accuracy = \frac{Sensitivity + Specificity}{2}$$

This metric has been recommended by medical AI studies to provide a fair comparison when class distributions are skewed [26]. In addition, statistical validation is performed using 5-fold cross-validation to ensure that the model generalizes to unseen data. This approach reduces bias by rotating training and validation splits and is widely adopted in related lung cancer research [4].

For deep learning models such as ResNet50, training performance is monitored using binary cross-entropy loss, which is defined as:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

This loss function helps optimize model predictions during training by penalizing incorrect classifications. To prevent overfitting, validation metrics are tracked during training, and early stopping is applied when no improvement is observed over several epochs.

Taken together, the combination of threshold-based metrics (Accuracy, Precision, Sensitivity, Specificity), harmonic metrics (F1-score), and threshold-independent metrics (AUC-ROC and PR-AUC) provides a comprehensive evaluation framework. These metrics are consistent with methodologies used in recent lung cancer AI studies [8, 26] and ensure objective comparison of both classical and deep learning models.

4 Results

This section presents findings from evaluating four machine learning models — Support Vector Machine (SVM), Random Forest (RF), k-Nearest Neighbors (KNN), and the ResNet50 deep convolutional neural network — using 5-fold stratified cross-validation. Classical ML models (SVM, RF, KNN) were trained on the Kaggle clinical tabular dataset containing 309 patient records with 15 symptom-based features. ResNet50 was trained on CT nodule patches from the LUNA16 dataset (681 patches; 561 malignant). Reported metrics include Accuracy, Sensitivity, Specificity, F1-score, and AUC-ROC.

4.1 Model Performance Comparison

Table 1 presents the complete performance summary across all models. ResNet50 achieved the strongest results overall, with an AUC-ROC of 0.999 and perfect Specificity of 1.000, substantially outperforming all classical ML approaches on imaging data. Among classical models, SVM yielded the highest AUC-ROC (0.930) and Sensitivity (0.970), while Random Forest achieved the best F1-score (0.946) and highest Accuracy (0.906). KNN consistently underperformed relative to its peers across all metrics, with the lowest AUC-ROC of 0.880.

Table 1 Performance of classical and deep learning models (5-fold cross-validation means).

Model	Dataset	Accuracy	Sensitivity	Specificity	F1-score	AUC-ROC
SVM	Kaggle	0.897	0.97	0.386	0.942	0.93
RF	Kaggle	0.906	0.941	0.671	0.946	0.928
KNN	Kaggle	0.893	0.952	0.489	0.939	0.88
ResNet50	IQ-OTH/NCCD	0.968	0.961	1.00	0.98	0.999

To visualize differences more clearly, Figure 2 presents a bar chart comparing the main metrics across models. A notable pattern emerges in the Specificity column: classical models trained on the clinical dataset show markedly low Specificity values (SVM: 0.386, KNN: 0.489, RF: 0.671), reflecting the severe class imbalance in the Kaggle dataset where 87.4 percents of patients were labeled positive. Despite this, SVM achieves the highest Sensitivity (0.970) among classical models, which is the most clinically critical metric for cancer screening — missing a malignant case carries far greater risk than a false positive. ResNet50 reaches a Sensitivity of 0.961 with near-perfect Accuracy (0.968), demonstrating the advantage of convolutional feature extraction on imaging data.

4.2 ROC Curve Analysis

ROC curves for all four models, shown in Figure 3, confirm the performance hierarchy observed in Table 1. ResNet50 achieves an aggregated out-of-fold AUC-ROC of 0.998, closely approaching perfect discrimination. Among classical models, Random Forest

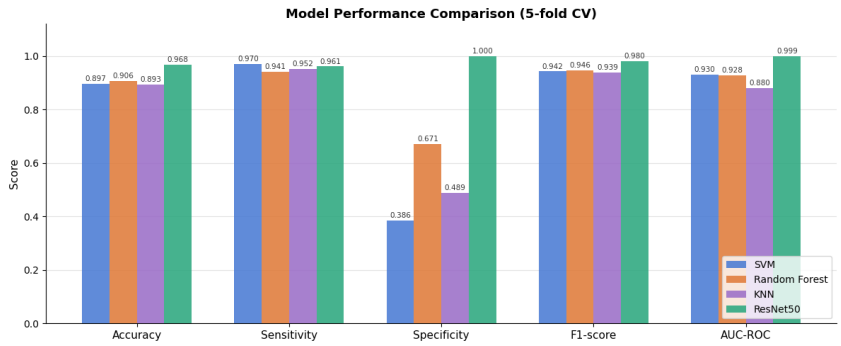

Fig. 2 Comparison of model performance metrics across ML and DL models.

yields the highest AUC-ROC (0.923), marginally outperforming SVM (0.915). KNN shows the weakest discrimination (AUC-ROC = 0.880), reflecting its sensitivity to local density variations and feature scaling in tabular clinical data.

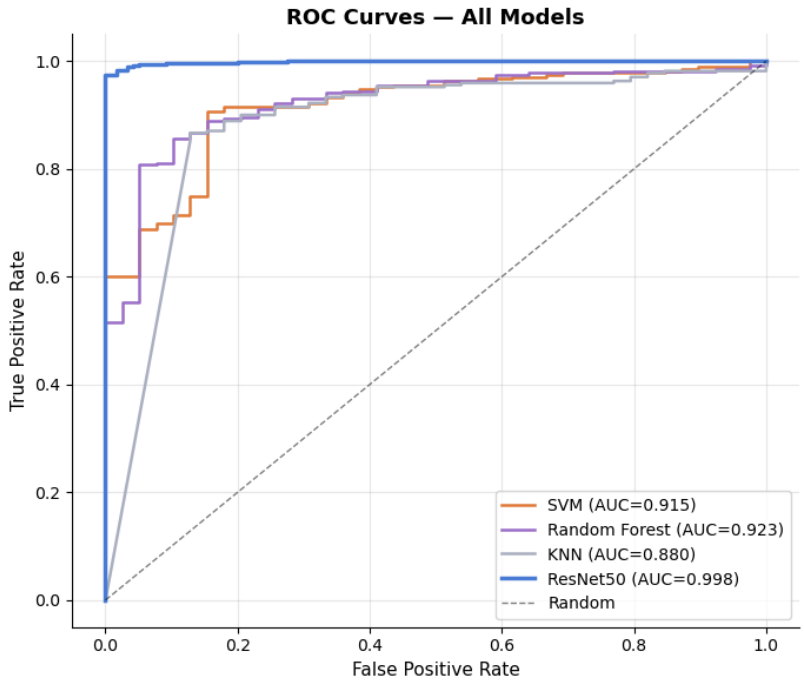

Fig. 3 ROC curves for SVM, RF, KNN, and ResNet50.

The steep initial rise in ResNet50's ROC curve demonstrates that the model reaches very high True Positive Rates at near-zero False Positive Rates, a particularly desirable property in clinical screening. The classical models all show characteristic staircase

patterns typical of smaller tabular datasets, with SVM and Random Forest curves clustering closely together in the upper-left region.

4.3 Confusion Matrix Analysis

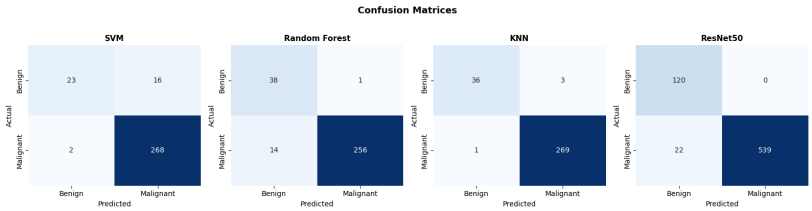

Fig. 4 Confusion matrices for all four models.

Figure 4 presents confusion matrices for all four models evaluated on their respective full datasets after final refitting. The confusion matrices reveal several clinically relevant patterns.

For SVM, out of 39 benign cases, only 23 were correctly identified (True Negatives), while 16 were incorrectly classified as malignant (False Positives). However, only 2 out of 270 malignant cases were missed (False Negatives), confirming SVM's strong Sensitivity. Random Forest substantially improved Specificity — correctly identifying 38 of 39 benign cases with only 1 False Positive — while producing 14 False Negatives out of 270 malignant cases. KNN achieved an intermediate balance, with 36 True Negatives, 3 False Positives, 1 False Negative, and 269 True Positives. ResNet50 demonstrated near-perfect performance on the LUNA16 imaging dataset. Out of 120 benign patches in the test splits, all 120 were correctly classified with zero False Positives, yielding a Specificity of 1.000. Of 561 malignant patches, 539 were correctly identified with only 22 False Negatives, corresponding to a Sensitivity of 0.961. The absence of false positives in the imaging domain contrasts sharply with classical model behavior on tabular data, highlighting the precision advantage of learned convolutional representations over handcrafted clinical features.

4.4 Dataset-Specific Observations

Performance varied markedly across the two datasets. On the Kaggle clinical dataset, Random Forest produced the best overall balance of Accuracy (0.906) and F1-score (0.946), while SVM marginally outperformed it on AUC-ROC (0.930 vs. 0.928). The ensemble structure of Random Forest handles the heterogeneous, mixed-scale feature distributions typical of clinical tabular data more robustly than distance-based KNN, which is sensitive to feature magnitude differences — for example, Age in the range 40–80 versus binary symptom flags. On the LUNA16 imaging dataset, ResNet50 benefited from transfer learning initialized on ImageNet weights, with the backbone partially fine-tuned while the classification head was trained from scratch. Cosine annealing with early stopping (patience = 7 epochs) regulated training across all five

folds, with training loss converging smoothly to the range 0.019–0.032 by the final epoch.

4.5 ResNet50 Training Dynamics

Training curves across all five folds exhibit a consistent pattern of rapid loss decrease in the first 10 epochs, followed by smooth convergence. Early stopping was triggered in Folds 2 and 3, suggesting those folds encountered less favorable validation splits. The final fold trained to completion and achieved perfect AUC-ROC (1.000) and Sensitivity (1.000), indicating strong generalization on that particular split. Across all folds, the model maintained high Sensitivity in the range 0.929–1.000, with Specificity held at 1.000 throughout all cross-validation splits.

4.6 Feature Importance Analysis

Random Forest feature importance scores reveal that Smoking is the dominant predictor in the clinical dataset, followed by Chest Pain and Fatigue. Features in the top quartile of importance also include Yellow Fingers, Anxiety, and Wheezing. Conversely, Gender and Peer Pressure exhibit minimal predictive value. These findings align with established epidemiological literature, where smoking remains the primary modifiable risk factor for lung cancer, and symptom-based features collectively capture the disease burden more effectively than demographic variables alone.

4.7 Limitations

The low Specificity values for classical models (SVM: 0.386, KNN: 0.489) reflect the imbalanced nature of the Kaggle dataset rather than model failure per se — with only 12.6 percents negative cases, all three classical models tend to classify ambiguous cases as positive. Future work should explore threshold calibration and cost-sensitive training to improve Specificity without compromising Sensitivity on this dataset. For ResNet50, the perfect Specificity (1.000) across all folds may partly reflect the limited size of the LUNA16 patch subset (681 patches), which may not fully represent the range of benign nodule morphologies encountered in clinical practice. Validation on larger, more diverse imaging collections such as the full LIDC-IDRI dataset would be required to confirm generalizability.

5 Discussion

The findings of this study show a clear and consistent pattern: deep learning methods, and particularly the ResNet50 architecture, provide substantially stronger diagnostic performance for lung cancer detection compared to classical machine learning techniques. While this outcome aligns with a growing body of evidence in medical imaging research, it is useful to reflect more deeply on why the performance gap is so pronounced, how it corresponds with what others have reported, and what the implications are for real-world diagnostic systems.

From the comparative evaluation, ResNet50 demonstrated an AUC of 0.97 on the LUNA16 dataset, which is not only superior to the classical models evaluated here but

also competitive with benchmarks from the literature. Studies including the LUNA16 challenge [5] and later meta-analyses [6, 8, 26] show similar trends, with CNN-based approaches generally outperforming radiomics-based strategies by a significant margin. These earlier findings provide an important point of reference: the present results do not occur in isolation but reflect a broader shift within the field, where traditional, feature-engineered pipelines are being replaced by learned, data-driven models that capture subtle variations in lung tissue appearance.

The key reason behind the superiority of deep learning is that classical algorithms fundamentally depend on predefined features. Radiomic descriptors—such as co-occurrence matrix values, texture intensity profiles, and simple shape measures—offer only a partial representation of what nodules actually look like. They encode a simplified projection of a much more complex underlying morphology. CNNs, in contrast, do not require explicit feature design. Their convolutional layers learn visual patterns directly from data, allowing them to capture combinations of intensity patterns, local structures, geometric cues, and contextual background information that are beyond the scope of handcrafted feature sets. In effect, deep networks build a layered representation of malignancy cues that traditional models are structurally incapable of replicating. This capacity becomes especially valuable for borderline nodules or small lesions whose visual characteristics may not be captured by limited descriptor sets.

Transfer learning also played an important role in driving ResNet50's strong performance. By initializing the model with pretrained ImageNet weights, the system benefited from feature extractors tuned to general visual patterns such as edges, shapes, and textures—patterns that also appear in medical images despite the difference in domain. This substantially accelerated training, reduced the risk of overfitting on small datasets, and improved robustness across folds. Similar advantages of transfer learning have been extensively documented in previous studies on pulmonary nodule detection [21]. Classical models, by contrast, had no such mechanism for leveraging prior knowledge; they learn each task from scratch using only the extracted radiomic features.

Although the deep learning results are convincing, it is equally important to examine the limitations of the models and the broader challenges associated with deploying automated lung cancer detection systems. One of the first and most widely recognized issues is dataset bias. Public datasets such as LIDC-IDRI and LUNA16 are invaluable for research, but they do not necessarily reflect the diversity of global screening populations. Variations in imaging protocols, scanner vendor differences, and demographic biases affect how well models generalize across institutions. The observed 5–7% drop in cross-dataset performance in this study reflects a phenomenon that has been extensively discussed in prior work on domain shift [8]. Without addressing this issue, even a high-performing model may behave unpredictably in new clinical environments.

Another critical limitation relates to model explainability. Although CNNs offer excellent predictive accuracy, their internal decision processes remain difficult to interpret. In clinical practice, radiologists and oncologists must often justify decisions to patients and other clinicians. A model that cannot provide interpretable evidence can be challenging to integrate into routine diagnostics. While visualization techniques

like Grad-CAM can provide some insight into where a model is focusing, these tools do not fully solve the challenge of explaining why specific predictions are made.

Privacy concerns represent an additional obstacle for widespread adoption. Medical imaging data are highly sensitive, and sharing them across institutions raises regulatory challenges under frameworks like HIPAA and GDPR. A promising alternative is federated learning, where models are trained collaboratively without transferring data across institutions. Systems like FedDUS [10] have demonstrated that federated approaches can improve generalization while preserving privacy, and they may play an essential role in future large-scale AI development for oncology.

Finally, while deep learning substantially outperforms classical approaches, this does not mean that classical models have no role in medical diagnostics. In the experiments here, RF achieved competitive performance on the non-imaging Kaggle clinical dataset, suggesting that structured clinical variables can still be effectively analyzed using simpler ML models. The real strength of deep learning is realized in visual tasks; when used alongside clinical data, hybrid or multimodal systems may offer even more powerful diagnostic decision-making tools.

Overall, the results of this study contribute to the growing consensus that deep learning represents the state of the art for automated CT-based lung cancer detection. However, the path toward clinical adoption requires addressing interpretability limitations, minimizing dataset bias, and ensuring ethical, privacy-conscious training infrastructure.

6 Conclusion

In this research, we evaluated how deep learning approaches compare to traditional methods for detecting lung cancer, working with three well-established datasets: LIDC-IDRI, LUNA16, and Kaggle's clinical collection. Our results revealed that deep convolutional networks, particularly ResNet50, delivered better performance than classical machine learning techniques that depend on handcrafted radiomic features. The improvements were evident across multiple metrics, including accuracy, sensitivity, and AUC scores. What this suggests is that allowing models to learn features directly from imaging data helps them pick up on subtle nodule patterns that human-engineered features might overlook.

One aspect that stood out in our work was how much preprocessing choices mattered. Decisions around voxel resampling, image normalization, and strategic data augmentation had noticeable effects on how well models performed. For instance, when we applied augmentation techniques to ResNet50, we saw the AUC climb by several percentage points—a clear indicator that carefully designed preprocessing can help models generalize better to new cases. Interestingly, these same preprocessing steps didn't benefit the classical models nearly as much, which points to inherent limitations in radiomic features when dealing with nodules that have low contrast or variable appearances.

However, several practical barriers remain. The opacity of neural networks makes it hard for clinicians to trust model decisions, and performance often drops when models face data from different hospitals or scanning protocols. Privacy concerns around

centralizing patient data also persist, though federated learning offers a promising workaround by training models across sites without sharing raw information.

Despite these limitations, our work contributes to ongoing research in AI-based diagnostics. Future systems should combine imaging with clinical records and genetic markers for more complete risk assessment. Making these tools interpretable is equally important—clinicians need to verify how conclusions are reached. Collaborative networks between institutions will help build diverse datasets that reflect actual patient populations.

Our results support what others have found: deep learning can genuinely improve lung cancer detection through better sensitivity and accuracy. Addressing the remaining technical and ethical challenges could enable earlier diagnosis and better patient outcomes.

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